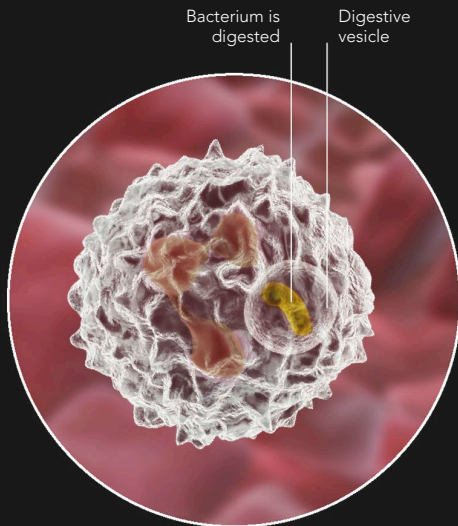
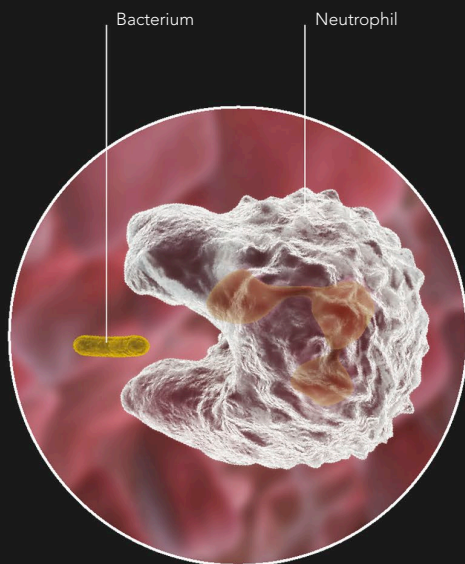


PHAGOCYTOSIS

Various kinds of white blood cell can surround, engulf, and ingest smaller items, such as bacteria and cellular debris, in a process known as phagocytosis ("cell eating"). The cell exploits its ability to change shape and move, using the intracellular components of microtubules and microfilaments (see p.26) that form its flexible, mobile, internal scaffolding. The ingestion usually takes less than one second, and the consumed material is gradually broken down by enzymes and other chemicals within the cell.

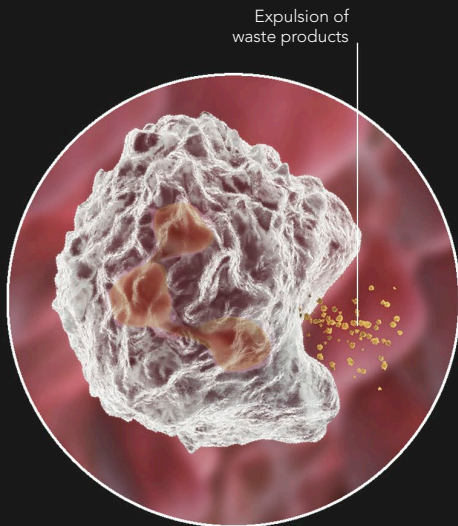
1 ENGULFING STAGE

The white cell extends pseudopods ("false feet") towards and around the unwanted item – here a bacterium. The pseudopods merge to engulf it.



2 LYSIS STAGE

Any unwanted items are trapped in phagocytic vesicles. Together with enzyme-containing lysosomes, these digestive vesicles form phagolysosomes, where lysis (breaking down) occurs.



3 EXOCYTOSIS STAGE

Harmless waste products of cell-eating are expelled through the membrane of the white blood cell, or in tiny, membrane-bound, exocytic vesicles, to the extracellular fluid.